

Backend Code

Student Management System

Node.js + Express REST API. Files: package.json, server.js.

API Endpoints

Method	Endpoint	Description
GET	/api/students?search=	List students (optional search)
POST	/api/students	Add a new student
PUT	/api/students/:id	Update a student
DELETE	/api/students/:id	Delete a student

package.json

```
{
  "name": "student-management-system",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "cors": "^2.8.5",
    "express": "^4.19.2",
    "mysql2": "^3.9.7"
  }
}
```

server.js

```
const express = require('express');
const cors = require('cors');
const mysql = require('mysql2');
const path = require('path');

const app = express();
app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));

// ---- MySQL connection pool (change password to your own) ----
const db = mysql.createPool({
  host: 'localhost',
  user: 'root',
  password: 'your_password',
  database: 'student_db'
});

// Common error handler
function fail(res, err) {
  if (err.code === 'ER_DUP_ENTRY')
    return res.status(409).json({ error: 'Email already exists' });
  res.status(500).json({ error: err.message });
}

// READ: list all students (optional ?search=)
app.get('/api/students', (req, res) => {
  const q = '%' + (req.query.search || '') + '%';
  db.query(
```

```

    'SELECT * FROM students WHERE name LIKE ? OR course LIKE ? ORDER BY id DESC',
    [q, q],
    (err, rows) => (err ? fail(res, err) : res.json(rows))
  );
});

// CREATE: add a student
app.post('/api/students', (req, res) => {
  const { name, email, course, year, phone } = req.body;
  if (!name || !email || !course || !year)
    return res.status(400).json({ error: 'Required fields missing' });
  db.query(
    'INSERT INTO students (name, email, course, year, phone) VALUES (?, ?, ?, ?, ?)',
    [name, email, course, year, phone || null],
    (err, result) =>
      err ? fail(res, err) : res.status(201).json({ id: result.insertId })
  );
});

// UPDATE: edit a student
app.put('/api/students/:id', (req, res) => {
  const { name, email, course, year, phone } = req.body;
  db.query(
    'UPDATE students SET name=?, email=?, course=?, year=?, phone=? WHERE id=?',
    [name, email, course, year, phone || null, req.params.id],
    (err, result) => {
      if (err) return fail(res, err);
      if (result.affectedRows === 0)
        return res.status(404).json({ error: 'Student not found' });
      res.json({ message: 'Updated' });
    }
  );
});

// DELETE: remove a student
app.delete('/api/students/:id', (req, res) => {
  db.query('DELETE FROM students WHERE id=?', [req.params.id], (err, result) => {
    if (err) return fail(res, err);
    if (result.affectedRows === 0)
      return res.status(404).json({ error: 'Student not found' });
    res.json({ message: 'Deleted' });
  });
});

const PORT = 3000;
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));

```